

5 marks

1. In each box mark T or F if the affirmation is true or false respectively. No justification is needed. Assume all functions below are continuous and that D is a bounded and closed fixed set.

☐ If $f(x, y) = g(x, y) - h(x, y)$ then

$$\iint_D f(x, y) \, dA = \iint_D g(x, y) \, dA - \iint_D h(x, y) \, dA.$$

☐ If $\iint_D f(x, y) \, dA = \iint_D g(x, y) \, dA$ then $f(x, y) = g(x, y)$ for each $(x, y) \in D$.

☐ If $\iint_D f(x, y) \, dA \geq 0$ then $f(x, y) \geq 0$ for each (x, y) in D .

☐ If $f(x, y) \geq 0$ in D then $\iint_D f(x, y) \, dA \geq 0$.

☐ If $\iint_D f(x, y) \, dA \leq 0$ then $f(x, y) \leq 0$ for some $(x, y) \in D$.

Solution: 1) True. 2) False. 3) False. 4) True, 5) True.

5 marks

2. Compute the volume of the solid on the first octant ($x, y, z \geq 0$) bounded by the planes $y = x$, $x = 1$ and $z = x + y$.

Solution: We want $I = \iint_D \frac{x+y}{2} \, dx \, dy$ where D is the region bounded by $y = x$, $y = 0$ and $x = 1$. Hence,

$$\begin{aligned} I &= \int_0^1 \int_y^1 (x + y) \, dx \, dy \\ &= \int_0^1 \left[\frac{x^2}{2} + yx \right]_{x=y}^{x=1} \, dy \\ &= \int_0^1 \left(\frac{1}{2} + y - \frac{3y^2}{2} \right) \, dy \\ &= \left[\frac{y}{2} + \frac{y^2}{2} - \frac{y^3}{2} \right]_{y=0}^{y=1} \\ &= \frac{1}{2}. \end{aligned}$$

5 marks

3. Compute the integral

$$I = \int_0^1 \int_y^1 e^{x^2} dx dy$$

Solution: Changing the order of integration we obtain:

$$I = \int_0^1 \int_0^x e^{x^2} dy dx$$

Hence,

$$I = \int_0^1 x e^{x^2} dx = \frac{1}{2} [e^{x^2}]_{x=0}^{x=1} = \frac{1}{2}(e - 1)$$

5 marks

4. Find the area between the curve
- $r = 2 + \sin(\theta)$
- and the circle of radius
- $r = 3$

Solution: Note that $2 + \sin(\theta) \leq 3$. We can thus write:

$$\begin{aligned} A &= \int_0^{2\pi} \int_{2+\sin(\theta)}^3 1 \cdot R dR d\theta \\ &= \int_0^{2\pi} \left[\frac{R^2}{2} \right]_{2+\sin(\theta)}^3 d\theta \\ &= \frac{1}{2} \int_0^{2\pi} 5 - 2\sin(\theta) - \sin(\theta)^2 d\theta \\ &= \frac{1}{2} \int_0^{2\pi} 5 - 2\sin(\theta) - \frac{1 - \cos(2\theta)}{2} d\theta \\ &= \frac{1}{2} \left[5\theta + 2\cos(\theta) - \frac{\theta}{2} + \frac{\sin(2\theta)}{4} \right]_0^{2\pi} \\ &= \frac{1}{2}(10\pi - \pi) = \frac{9\pi}{2} \end{aligned}$$

2 marks

5. (Bonus marks) Find the volume enclosed by the cylinders
- $z = x^2 + y^2$
- ,
- $y = x^2 + z^2$
- and
- $x = y^2 + z^2$
- .

Solution: $V = 16 - 8\sqrt{2}$. See Midterm2 2018 Summer term for the full explanation.